Shuhei Watanabe

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Education

Oct 2020 – Oct 2023	**University of Freiburg**, Freiburg, Germany. MSc in Computer Science. Supervisor: Prof. Frank Hutter. Overall GPA: 1.1/5.0 (1.0 is the best grade). Thesis: Significant Runtime Reduction for Asynchronous Multi-Fidelity Optimization on Zero-Cost Benchmarks.
Apr 2016 – Mar 2020	**The University of Tokyo**, Tokyo, Japan. BSc in Systems Innovation. **Ranked 1st** out of 37 students by GPA.
Apr 2014 – Mar 2016	**The University of Tokyo**, Tokyo, Japan. Bachelor of College of Arts and Science, Natural Science 1.

Employment

Jan 2026 – Present	**SB Intuitions Corp.**, Tokyo, Japan. (Robotics Senior Research Scientist) Involved in cloth folding demo project.
Jun 2024 – Dec 2025	**National Institute of Advanced Industrial Science and Technology (AIST)**, Tokyo, Japan. (Part Time Visiting Researcher at Social Intelligence Research (SIR) Team)
Oct 2023 – Dec 2025	**Preferred Networks Inc.**, Tokyo, Japan. (Full Time Research Engineer) Core Optuna developer. Delivered significant speedup of `TPESampler` (300x), the default sampler in Optuna, and `GPSampler` (10x faster than BoTorch). Led the development of `GPSampler` and its extensions. Deployed Optuna in materials & physics simulation applications.
Dec 2020 – Sep 2023	**Machine Learning Lab**, Freiburg, Germany. Development of Auto-PyTorch, an AutoML tool.
Sep 2018 – Sep 2020	**AIST**, Tokyo, Japan. (Full Time Technical Staff at SIR Team) Co-first authored "Warm Starting CMA-ES for Hyperparameter Optimization" (AAAI'21). Conducted a large-scale experiment on a cluster.

Awards / Honors

Aug 2026	**KDD Best Reviewer** (Research Track), ACM SIGKDD 2026.
Sep 2023	**AutoML 2023 Travel Awards** (500 EUR).
Aug 2023	**IJCAI-AIJ 2023 Travel and Accessibility Grant Program** (1,000 USD).
Oct 2022	**NeurIPS 2022 Complimentary Registration** (350 USD).
Oct 2022	**ELIZA MSc Scholarship** (12,000 EUR, 4 students selected from the University).
Oct 2022	**Deutschlandstipendium** (3,600 EUR).
Jul 2022	**1st Prize in AutoML2022: Multiobjective Hyperparameter Optimization for Transformers**

Sep 2020 **ITO Foundation for International Education Exchange** (48,000 USD, AR: 6.7%).

Mar 2020 **Hatakeyama Award from the Japan Society of Mechanical Engineers**. Awarded for the distinctive grades at the University of Tokyo (AR: 5/340=1.5%).

May 2019 **PRMU 2018 Yearly Research Encouragement Award**. For "*Speed Up of Hyper-Parameter Tuning with Nelder-Mead Method by Parallel Computing*" (AR: 3/170=1.8%).

Publications & Talks

AR, ◯, and ♣ denote the acceptance rate, the presenter, and equal contribution, respectively. The total citations on Google Scholar is **1800+** as of October 2026.

Refereed Conference Publications

1. ◯ K. Baba, Y. Ozaki, **S. Watanabe** (2026). Conditional PED-ANOVA: Hyperparameter Importance in Hierarchical & Dynamic Search Spaces. ACM SIGKDD Conference on Knowledge Discovery & Data Mining (KDD).
2. ◯ C. Mori, **S. Watanabe**, M. Onishi, T. Itoh (2025). User Preference-Based Parallel Coordinate Plots: Its Application in Guidance Planning. International Conference on Pedestrian and Evacuation Dynamics (PED).
3. ◯ ♣ C. Mori, ♣ **S. Watanabe**, M. Onishi, T. Itoh (2025). Preference-Optimal Multi-Metric Weighting for Parallel Coordinate Plots. International Conference Information Visualisation (iV).
4. ◯ **S. Watanabe**, N. Mallik, E. Bergman, F. Hutter (2024). Fast Benchmarking of Asynchronous Multi-Fidelity Optimization on Zero-Cost Benchmarks. AutoML Conference. Preliminary: AutoML Conference Workshop Track 2023.
5. ◯ **S. Watanabe**, F. Hutter (2023). c-TPE: Tree-Structured Parzen Estimator with Inequality Constraints for Expensive Hyperparameter Optimization. International Joint Conference on Artificial Intelligence (IJCAI) (AR: 644/4566≃14%). Preliminary: GPSMDM Workshop at NeurIPS 2022.
6. ◯ **S. Watanabe**, N. Awad, M. Onishi, F. Hutter (2023). Speeding Up Multi-Objective Hyperparameter Optimization by Task Similarity-Based Meta-Learning for the Tree-Structured Parzen Estimator. International Joint Conference on Artificial Intelligence (IJCAI) (AR: 644/4566≃14%). Preliminary: MetaLearn Workshop at NeurIPS 2022.
7. ◯ **S. Watanabe**, A. Bansal, F. Hutter (2023). PED-ANOVA: Efficiently Quantifying Hyperparameter Importance in Arbitrary Subspaces. International Joint Conference on Artificial Intelligence (IJCAI) (AR: 644/4566≃14%). (**The default algorithm in Optuna**)
8. ◯ S. Shigenaka, S. Takami, **S. Watanabe**, Y. Tanigaki, Y. Ozaki, M. Onishi (2021). MAS-Bench: Parameter Optimization Benchmark for Multi-Agent Crowd Simulation. International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS).
9. ◯ ♣ M. Nomura, ♣ **S. Watanabe**, Y. Akimoto, Y. Ozaki, M. Onishi (2021). Warm Starting CMA-ES for Hyperparameter Optimization. AAAI Conference on Artificial Intelligence (AAAI). (AR: 1692/9034=19%). Preliminary: MetaLearn Workshop at NeurIPS 2019.

10. ◯ S. Takenaga, **S. Watanabe**, M. Nomura, Y. Ozaki, M. Onishi, H. Habe (2020). Evaluating Initialization of Nelder–Mead Method for Hyperparameter Optimization in Deep Learning. International Conference on Pattern Recognition (ICPR). Oral Presentation.
11. ◯ Y. Ozaki, Y. Tanigaki, **S. Watanabe**, M. Onishi (2020). Multiobjective Tree-Structured Parzen Estimator for Computationally Expensive Optimization Problems. The Genetic and Evolutionary Computation Conference (GECCO).
12. ◯ **S. Watanabe**, Y. Ozaki, Y. Bando, M. Onishi (2019). Speeding Up of the Nelder–Mead Method by Data–Driven Speculative Execution. Asian Conference on Pattern Recognition (ACPR). Oral Presentation. (**Oral presentation: 36/273=13%**)

Refereed Journal Publications

1. ♣ Y. Ozaki, ♣ **S. Watanabe**, T. Yanase (2026). OptunaHub: A Platform for Black-Box Optimization. Journal of Machine Learning Research (JMLR).
2. ◯ S. Shigenaka, S. Takami, **S. Watanabe**, Y. Tanigaki, M. Onishi (2024). MAS-Bench: A Benchmarking for Parameter Calibration of Multi-Agent Crowd Simulation. Journal of Computational Social Science.
3. ◯ Y. Ozaki, Y. Tanigaki, **S. Watanabe**, M. Nomura, M. Onishi (2022). Multiobjective Tree-Structured Parzen Estimator. Journal of Artificial Intelligence Research (JAIR).

Selected Non-Archival Publications

1. ◯ ♣ **S. Watanabe**, ♣ K. Baba (2027). No Sigmoid Required: Exact Feasibility for Constrained qEHVI. Under review at International Conference on Learning Representations (ICLR).
2. ◯ K. Irie, **S. Watanabe**, M. Onishi (2026). Batch Acquisition Function Evaluations and Decouple Optimizer Updates for Faster Bayesian Optimization. Workshop on AI to Accelerate Science and Engineering (AI2ASE) at AAAI.
3. **S. Watanabe** (2023). Tree-Structured Parzen Estimator: Understanding Its Algorithm Components and Their Roles for Better Empirical Performance. arXiv:2304.11127. (**900+ citations on Google Scholar**, the first detailed paper about the Optuna default sampler)
4. ◯ Y. Ozaki, ◯ **S. Watanabe**, M. Onishi (2019). Accelerating the Nelder–Mead Method with Predictive Evaluation. Workshop on Automated Machine Learning (AutoML) at ICML.
5. ◯ **S. Watanabe**, Y. Ozaki, M. Onishi (2019). Speed Up of Hyper-Parameter Tuning with Nelder–Mead Method by Parallel Computing. Pattern Recognition and Media Understanding (PRMU). **PRMU 2018 Yearly Research Encouragement Award** (AR: 3/170=1.8%).

Talks

1. ◯ **S. Watanabe** (2026). LLM Meets Optuna: Potential, Challenges, and Lessons. Meta Adaptive Experimentation Workshop. (Invited)
2. ◯ ♣ **S. Watanabe**, ◯ ♣ Y. Ozaki (2025). Future Directions of AutoML from the Optuna View. NII Shonan Meeting (Advancing Automated Machine Learning). (Invited)

3. ◯ **S. Watanabe**, H. Imamura, C. Shinagawa, K. Shinohara, S. Takamoto, J. Li (2024). Multi-Objective Bayesian Optimization for Materials Discovery with Neural Network Potential – An Application to Li-Ion Battery Cathode Material. Materials Research Society Fall Meeting & Exhibit.

Academic Service

Reviewer

- International Conference on Learning Representations (2027).
- ACM SIGKDD Conference on Knowledge Discovery & Data Mining (2026). Recognized as a **KDD Best Reviewer** (Research Track).
- AutoML Conference (2023).

Mentoring & Supervision

Jun 2024 – Mar 2026	**Chisa Mori**, MSc Student, AIST. Parallel coordinate plots for multi-objective problems (Accepted to iV).
Jul 2024 – Present	**Kaito Baba**, MSc Student, Preferred Networks Inc. 1. Development of constrained optimization for the Gaussian process-based sampler (single-objective, multi-objective). 2. PED-ANOVA for conditional search spaces (Accepted to KDD).
Aug 2025 – Present	**Kaichi Irie**, MSc Student, Preferred Networks Inc. & AIST. Development of parallel processing in the Gaussian process-based sampler (article, workshop paper).
Jun 2026 – Sep 2026	**Eri Sawada**, MSc Student, Preferred Networks. Development of batch suggestion in the Gaussian process-based sampler (article).

Technical Highlights

- Coding-Agent Management (token efficiency tracking)
- Applications of Bayesian Optimization (materials science & engineering design)
- Physics-Based Simulation (lattice Boltzmann, numerical integration, constraint satisfaction)
- Cluster Experience (MPI, parallel programming, large-scale experiments, MOAB, Slurm)
- Software Engineering (Python, C++, PyTorch, team development)

Miscellaneous

- **Japanese** (Native Language), **English** (C1, TOEFL iBT: 100), **German** (B2)
- Approx. Top 3.5% (highest) algorithm programmer in **AtCoder** mostly using C++.